Chapter 8: Probability

Student Question Booklet

Sections 8A–8J : Year 10 Mathematical Methods : 2026

How to use this booklet

This booklet is your **map through sections 8A–8J** — Venn diagrams and two-way tables, the addition rule, conditional probability, arrays, tree diagrams, independence and, for 10A, counting arrangements and selections. For every section you will find:

- **Worked examples** — fully solved examples with a diagram where it helps, plus the matching textbook example numbers and pages to cross-reference.
- **Question sets** — the questions to practise, grouped by *Building Understanding*, *Fluency*, *Problem-solving*, *Reasoning* and *Enrichment*, with page numbers. Find them in your own textbook.
- **Hurdle questions** — five questions you **must** be able to do to pass the section, with working space (and a diagram wherever you need to draw or complete one).

Diagrams are schematic (not to scale). Sections marked 10A are extension. Give probabilities as fractions in simplest form, or as decimals when the question uses decimals. Always show working — write the rule or draw the diagram before you calculate.

Page references are to *Essential Mathematics for the Victorian Curriculum 10 & 10A*, 3rd ed. Question sets belong to the textbook; the worked examples, diagrams and hurdle questions are original material written for this course. Section 8K is not part of this course.

8A Review of probability CONSOLIDATING

Textbook worked examples (cross-reference in your textbook)

- **Example 1** — Calculating simple theoretical probabilities (p. 684)
- **Example 2** — Calculating simple experimental probabilities (p. 684)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 683)
- **Fluency:** Q1–5 (p. 685–686)
- **Problem-solving:** Q6–9 (p. 686–687)
- **Reasoning:** Q10–12 (p. 687)
- **Enrichment: Target probability:** Q13 (p. 687)

Worked examples (study the method, then try the hurdles)

Theoretical probability from a spinner

The spinner below has eight equal sectors. Find:

(a) Pr(3)

(b) Pr(an even number)

(c) Pr(not 3)

(d) Pr(a number greater than 3).



Solution. All eight sectors are equally likely, so

$$\Pr(\text{event}) = \frac{\text{number of favourable sectors}}{8}.$$

(a) Three sectors show 3: $\Pr(3) = \frac{3}{8}$.

(b) The even sectors are $2, 2, 4$: $\Pr(\text{even}) = \frac{3}{8}$.

(c) $\Pr(\text{not } 3) = 1 - \Pr(3) = 1 - \frac{3}{8} = \frac{5}{8}$.

(d) The sectors 4 and 5 are greater than 3: $\Pr(\text{greater than } 3) = \frac{2}{8} = \frac{1}{4}$.

Experimental probability from a frequency table

A die is rolled 200 times to test whether it is fair.

Outcome	1	2	3	4	5	6
Frequency	28	31	30	33	26	52

(a) Find the experimental probability of rolling a 6.

(b) Find the experimental probability of rolling an odd number.

(c) Does the die appear to be fair? Explain.

(d) Using these results, how many 6s would you expect in 500 rolls?

Solution. Experimental probability is found from the results:

$$\text{experimental probability} = \frac{\text{number of times the event happened}}{\text{total number of trials}}.$$

(a) $\Pr(6) = \dfrac{52}{200} = 0.26$.

(b) Odd numbers: $28 + 30 + 26 = 84$, so $\Pr(\text{odd}) = \dfrac{84}{200} = 0.42$.

(c) For a fair die, $\Pr(6) = \dfrac{1}{6} \approx 0.17$. Over 200 rolls we would expect about 33 sixes, not 52, so the die appears to be **biased towards 6**.

(d) $0.26 \times 500 = 130$ sixes.

Notes

Hurdle questions — Section 8A (show all working)

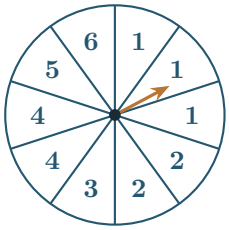
H1. A letter is chosen at random from the word **MISSISSIPPI**. Find:

(a) Pr(S)

(b) Pr(a vowel)

(c) Pr(not P).

H2. The spinner below has ten equal sectors.



(a) Find Pr(1).

(b) Find Pr(an even number).

(c) Find Pr(a number less than 4).

(d) How many 4s would you expect in 50 spins?

H3. A traffic survey counted the passengers (not including the driver) in 120 cars.

Number of passengers	0	1	2	3	4
Number of cars	54	38	17	8	3

(a) Find the experimental probability that a car has exactly one passenger.

(b) Find the experimental probability that a car has at least two passengers.

(c) Estimate how many of the next 600 cars will have no passengers.

H4. A bag contains only red, blue and green counters. A counter is chosen at random. Pr(red) = 0.35 and Pr(blue) = 0.4.

(a) Find Pr(green).

(b) Find Pr(not blue).

(c) The bag holds 60 counters. How many of each colour are there?

H5. A fair 12-sided die, numbered 1 to 12, is rolled once. Find:

(a) Pr(a prime number)

(b) Pr(a multiple of 3 or a multiple of 4)

(c) Pr(a number greater than 12), and describe this event in words.

8B Set notation in Venn diagrams and two-way tables CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 3** — Listing sets (p. 690)
- **Example 4** — Using Venn diagrams (p. 691)
- **Example 5** — Using two-way tables (p. 692)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 689)
- **Fluency:** Q1–6 (p. 693–694)
- **Problem-solving:** Q7–10 (p. 694–695)
- **Reasoning:** Q11–13 (p. 695)
- **Enrichment: Triple Venn diagrams:** Q14–15 (p. 696)

Worked examples (study the method, then try the hurdles)

Listing sets and drawing the Venn diagram

Let $\xi = \{1, 2, 3, \ldots, 12\}$, $A = \{\text{multiples of } 3\}$ and $B = \{\text{even numbers}\}$.

(a) List $A \cap B$, $A \cup B$ and A'.

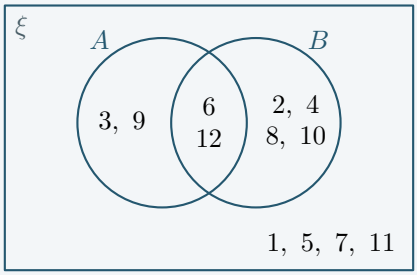
(b) Show the elements on a Venn diagram.

(c) A number is chosen at random from ξ. Find $\Pr(A \cup B)$.

Solution.

(a) $A = \{3, 6, 9, 12\}$ and $B = \{2, 4, 6, 8, 10, 12\}$, so
$A \cap B$ (in A *and* B) $= \{6, 12\}$
$A \cup B$ (in A *or* B or both) $= \{2, 3, 4, 6, 8, 9, 10, 12\}$
A' (not in A) $= \{1, 2, 4, 5, 7, 8, 10, 11\}$.

(b) Fill the overlap first, then the rest of each circle, then the numbers in neither set.

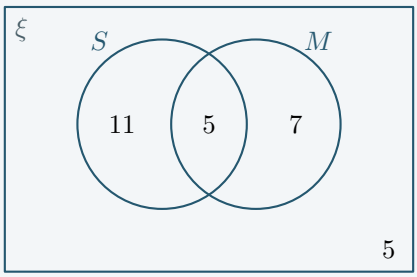


(c) $n(A \cup B) = 8$, so $\Pr(A \cup B) = \dfrac{8}{12} = \dfrac{2}{3}$.

Notes

From a Venn diagram to a two-way table

In a class of 28 students, 16 play a team sport (S), 12 play a musical instrument (M) and 5 do both. Draw a Venn diagram and a two-way table, then find the probability that a randomly chosen student does exactly one of the two activities.



	M	M'	**Total**
S	5	11	16
S'	7	5	12
Total	12	16	28

Solution. Start with the overlap: 5. Then sport only $= 16-5 = 11$, instrument only $= 12-5 = 7$ and neither $= 28-(11+5+7) = 5$. In the table, each row and column must add to its total. Exactly one activity means *sport only* or *instrument only*: $\Pr(\text{exactly one}) = \dfrac{11+7}{28} = \dfrac{18}{28} = \dfrac{9}{14}$.

Notes

Hurdle questions — Section 8B (show all working)

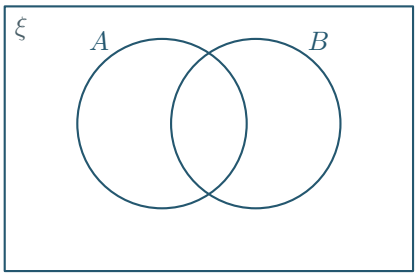
H1. Let $\xi = \{1, 2, 3, \ldots, 15\}$, $A = \{\text{factors of } 12\}$ and $B = \{\text{odd numbers}\}$.

(a) List the elements of A, $A \cap B$ and B'.

(b) Complete the Venn diagram.

(c) Find $n(A \cup B)$.

(d) A number is chosen at random from ξ. Find $\Pr(A \cap B')$.



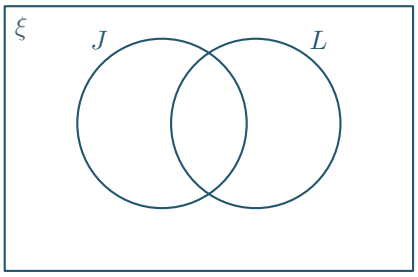
H2. Of 40 Year 10 students, 24 have a part-time job (J), 18 have a learner permit (L) and 9 have both.

(a) Complete the Venn diagram.

(b) Find Pr(learner permit only).

(c) Find $\Pr(J \cup L)$.

(d) Find Pr(neither).



H3. Sixty people were asked whether they own a cat (C) and whether they own a dog (D).

(a) Complete the two-way table.

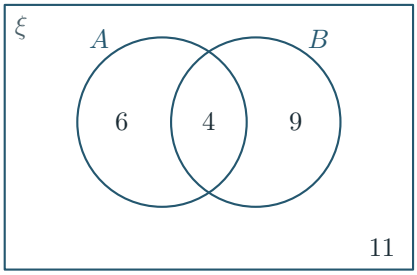
(b) Find $\Pr(C \cap D')$.

(c) Find $\Pr(D)$.

(d) Find $\Pr(C' \cap D')$.

	D	D'	**Total**
C	8		20
C'			
Total	26		60

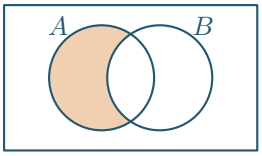
H4. The Venn diagram shows the number of elements in events A and B.



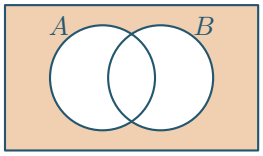
(a) Show the same information in a two-way table.

(b) Find $\Pr(A')$, $\Pr(A \cap B')$ and $\Pr(A \cup B)$.

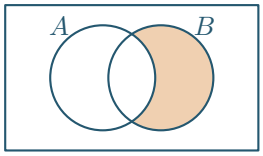
H5. Write the shaded region of each Venn diagram in set notation.



(a) _____



(b) _____



(c) _____

8C The addition rule CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 6** — Applying the addition rule (p. 698)
- **Example 7** — Using the addition rule (p. 700)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 698)
- **Fluency:** Q1–5 (p. 700–701)
- **Problem-solving:** Q6–9 (p. 701–702)
- **Reasoning:** Q10–12 (p. 702)
- **Enrichment: Divisibility and the addition rule:** Q13–14 (p. 702)

Worked examples (study the method, then try the hurdles)

The addition rule with counting

A number is chosen at random from 1 to 30. Let A be "a multiple of 4" and B be "a multiple of 6". Use the addition rule to find $\Pr(A \cup B)$, then check by listing.

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

Solution. $A = \{4, 8, 12, 16, 20, 24, 28\}$, so $\Pr(A) = \dfrac{7}{30}$.
$B = \{6, 12, 18, 24, 30\}$, so $\Pr(B) = \dfrac{5}{30}$.
The numbers in both are the multiples of 12: $A \cap B = \{12, 24\}$, so $\Pr(A \cap B) = \dfrac{2}{30}$.

$$\Pr(A \cup B) = \frac{7}{30} + \frac{5}{30} - \frac{2}{30} = \frac{10}{30} = \frac{1}{3}.$$

Check: $A \cup B = \{4, 6, 8, 12, 16, 18, 20, 24, 28, 30\}$ has 10 elements. We subtract $\Pr(A \cap B)$ because 12 and 24 were counted once in A and again in B.

Notes

The addition rule with given probabilities

For two events, $\Pr(A) = 0.45$, $\Pr(B) = 0.3$ and $\Pr(A \cup B) = 0.6$.

(a) Find $\Pr(A \cap B)$.

(b) Find $\Pr(A \cap B')$.

(c) Find $\Pr\left((A \cup B)'\right)$.

(d) Show all four regions on a Venn diagram.

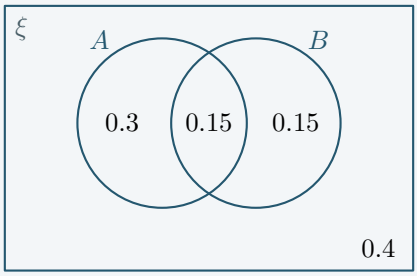
Solution.

(a) Rearrange the addition rule: $\Pr(A \cap B) = \Pr(A) + \Pr(B) - \Pr(A \cup B) = 0.45 + 0.3 - 0.6 = 0.15$.

(b) A only: $\Pr(A \cap B') = \Pr(A) - \Pr(A \cap B) = 0.45 - 0.15 = 0.3$.

(c) Neither: $\Pr\left((A \cup B)'\right) = 1 - \Pr(A \cup B) = 1 - 0.6 = 0.4$.

(d) B only is $0.3 - 0.15 = 0.15$. Check: $0.3 + 0.15 + 0.15 + 0.4 = 1$.



Mutually exclusive events have $\Pr(A \cap B) = 0$, and then $\Pr(A \cup B) = \Pr(A) + \Pr(B)$.

Notes

Hurdle questions — Section 8C (show all working)

H1. A number is chosen at random from 1 to 40. Let A be "a multiple of 5" and B be "a multiple of 8".

(a) Find $\Pr(A)$, $\Pr(B)$ and $\Pr(A \cap B)$.

(b) Use the addition rule to find $\Pr(A \cup B)$.

H2. For two events, $\Pr(A) = 0.6$, $\Pr(B) = 0.5$ and $\Pr(A \cap B) = 0.25$.

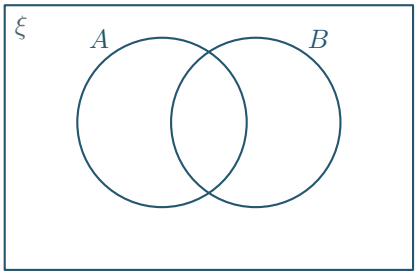
(a) Complete the Venn diagram with probabilities.

(b) Find $\Pr(A \cup B)$.

(c) Find $\Pr(A')$.

(d) Find $\Pr(A \cap B')$.

(e) Find the probability that neither event occurs.



H3. For two events, $\Pr(A) = 0.35$, $\Pr(B) = 0.4$ and $\Pr(A \cup B) = 0.75$. Are A and B mutually exclusive? Use the addition rule to explain.

H4. In a group of 50 students, 22 study French, 19 study Japanese and 8 study both. A student is chosen at random. Find the probability that the student studies:

(a) French or Japanese

(b) neither language

(c) exactly one of the two languages.

H5. For two events, $\Pr(A) = 0.7$ and $\Pr(B) = 0.5$. A student writes $\Pr(A \cup B) = 0.7 + 0.5 = 1.2$.

(a) Explain why this answer must be wrong, and what it tells you about A and B.

(b) If in fact $\Pr(A \cup B) = 0.9$, find $\Pr(A \cap B)$.

8D Conditional probability CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 8** — Finding conditional probabilities using a Venn diagram (p. 704)
- **Example 9** — Finding conditional probabilities using a two-way table (p. 705)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 704)
- **Fluency:** Q1–4 (p. 706–707)
- **Problem-solving:** Q5–8 (p. 707)
- **Reasoning:** Q9–11 (p. 708)
- **Enrichment: Investment workshops:** Q12 (p. 708)

Worked examples (study the method, then try the hurdles)

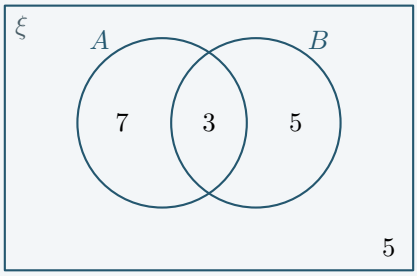
Conditional probability from a Venn diagram

The Venn diagram shows the number of elements in events A and B. Find:

(a) $\Pr(A \mid B)$

(b) $\Pr(B \mid A)$

(c) $\Pr(A)$.



Solution. $\Pr(A \mid B)$ means "the probability of A, *given* that B has happened". Knowing B happened shrinks the sample space to the circle B.

(a) $n(B) = 3 + 5 = 8$, and 3 of those are also in A: $\Pr(A \mid B) = \dfrac{n(A \cap B)}{n(B)} = \dfrac{3}{8}$.

(b) $n(A) = 7 + 3 = 10$: $\Pr(B \mid A) = \dfrac{n(A \cap B)}{n(A)} = \dfrac{3}{10}$.

(c) There are $7 + 3 + 5 + 5 = 20$ elements in total: $\Pr(A) = \dfrac{10}{20} = \dfrac{1}{2}$.

Note that $\Pr(A \mid B) \neq \Pr(B \mid A)$: the "given" event decides the denominator.

Notes

Conditional probability from a two-way table

A survey of 120 students recorded whether each wears glasses (G) and whether each is left-handed (L).

	L	L'	**Total**
G	6	30	36
G'	10	74	84
Total	16	104	120

Find:

(a) $\Pr(G \mid L)$

(b) $\Pr(L \mid G)$.

Solution.

(a) Given L: use the L *column* only (16 students), of whom 6 wear glasses: $\Pr(G \mid L) = \dfrac{6}{16} = \dfrac{3}{8}$.
The rule gives the same result:

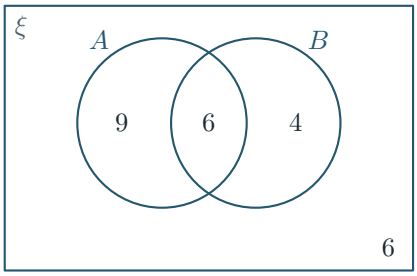
$$\Pr(G \mid L) = \frac{\Pr(G \cap L)}{\Pr(L)} = \frac{6/120}{16/120} = \frac{6}{16} = \frac{3}{8}.$$

(b) Given G: use the G *row* only (36 students): $\Pr(L \mid G) = \dfrac{6}{36} = \dfrac{1}{6}$.

Notes

Hurdle questions — Section 8D (show all working)

H1. The Venn diagram shows the number of elements in events A and B.



Find:

(a) $\Pr(A \mid B)$

(b) $\Pr(B \mid A)$

(c) $\Pr(A' \mid B)$

(d) $\Pr(B \mid A')$.

H2. Eighty people were asked whether they like coffee (C) and whether they like tea (T).

	T	T'	**Total**
C	18	34	52
C'	20	8	28
Total	38	42	80

Find:

(a) $\Pr(C \mid T)$

(b) $\Pr(T \mid C)$

(c) $\Pr(C' \mid T')$.

H3. For two events, $\Pr(A) = 0.5$, $\Pr(B) = 0.4$ and $\Pr(A \cap B) = 0.1$. Use $\Pr(A \mid B) = \dfrac{\Pr(A \cap B)}{\Pr(B)}$ to find:

(a) $\Pr(A \mid B)$

(b) $\Pr(B \mid A)$.

H4. For two events, $\Pr(A \mid B) = 0.3$ and $\Pr(B) = 0.6$.

(a) Find $\Pr(A \cap B)$.

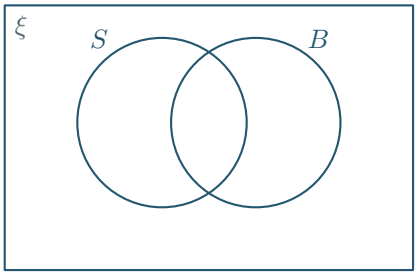
(b) If $\Pr(A) = 0.45$, find $\Pr(B \mid A)$.

H5. In a group of 30 students, 12 play soccer (S), 15 play basketball (B) and 4 play both.

(a) Complete the Venn diagram.

(b) A student chosen at random plays basketball. Find the probability that they also play soccer.

(c) A student chosen at random plays soccer. Find the probability that they also play basketball.



8E Two-step experiments using arrays CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 10** — Constructing a table with replacement (p. 710)
- **Example 11** — Constructing a table without replacement (p. 712)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 710)
- **Fluency:** Q1–4 (p. 713)
- **Problem-solving:** Q5–7 (p. 714)
- **Reasoning:** Q8–10 (p. 714–715)
- **Enrichment: Random weights:** Q11 (p. 715)

Worked examples (study the method, then try the hurdles)

An array with replacement

Two fair four-sided dice, each numbered 1 to 4, are rolled and the numbers are added. Draw an array of the totals, then find:

(a) Pr(total of 5)

(b) Pr(total of 7 or more)

(c) Pr(a double).

1st ↓ / 2nd →	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

Solution. Each die can show any number on either roll, so the array has $4 \times 4 = 16$ equally likely outcomes.

(a) A total of 5 appears 4 times (shaded): $\Pr(5) = \dfrac{4}{16} = \dfrac{1}{4}$.

(b) Totals of 7 or more are $7, 7, 8$: $\Pr(7 \text{ or more}) = \dfrac{3}{16}$.

(c) The doubles lie on the diagonal ($1{+}1, 2{+}2, 3{+}3, 4{+}4$): $\Pr(\text{double}) = \dfrac{4}{16} = \dfrac{1}{4}$.

Notes

An array without replacement

Five cards numbered 1 to 5 are shuffled. Two cards are drawn, one after the other, **without replacement**. Find:

(a) Pr(both even)

(b) Pr(the two cards add to 6)

(c) Pr(at least one card is a 5).

1st ↓ / 2nd →	1	2	3	4	5
1		1,2	1,3	1,4	1,5
2	2,1		2,3	2,4	2,5
3	3,1	3,2		3,4	3,5
4	4,1	4,2	4,3		4,5
5	5,1	5,2	5,3	5,4	

Solution. The first card cannot be drawn again, so the diagonal (the same card twice) is crossed out, leaving $5 \times 4 = 20$ outcomes.

(a) Both even: $(2,4)$ and $(4,2)$, so $\Pr(\text{both even}) = \dfrac{2}{20} = \dfrac{1}{10}$.

(b) Total of 6 (shaded): $(1,5), (5,1), (2,4), (4,2)$ — but *not* $(3,3)$ — so $\Pr = \dfrac{4}{20} = \dfrac{1}{5}$.

(c) At least one 5: the last row and the last column, 8 outcomes, so $\Pr = \dfrac{8}{20} = \dfrac{2}{5}$.

Notes

Hurdle questions — Section 8E (show all working)

H1. A coin is tossed and a fair six-sided die is rolled.

(a) Complete the array of outcomes (for example, H3).

(b) Find Pr(a head and an even number).

(c) Find Pr(a tail and a number greater than 4).

(d) Find Pr(a head or a 6).

1st ↓ / 2nd →	1	2	3	4	5	6
H						
T						

H2. Two fair six-sided dice are rolled and the numbers are added.

(a) Complete the array of totals.

(b) Find Pr(total of 7).

(c) Find Pr(total of 10 or more).

(d) Find Pr(the total is a prime number).

1st ↓ / 2nd →	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

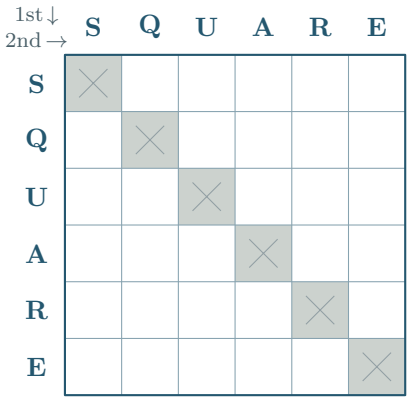
H3. Two letters are chosen at random, one after the other, **without replacement**, from the word **SQUARE**.

(a) Complete the array. How many outcomes are there?

(b) Find Pr(both letters are vowels).

(c) Find Pr(at least one letter is a vowel).

(d) Find Pr(the first letter is S).



H4. Two digits are chosen from $\{1, 2, 3\}$ to make a two-digit number (the first digit chosen is the tens digit).

(a) The digits are chosen **with** replacement.
List all the possible numbers, then find Pr(greater than 20) and Pr(even).

(b) The digits are chosen **without** replacement.
List all the possible numbers, then find Pr(even).

H5. Explain why the diagonal of an array is crossed out when items are selected **without** replacement. Then find the number of outcomes when two items are chosen in order from 8 different items:

(a) with replacement

(b) without replacement.

8F Using tree diagrams CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 12** — Constructing a tree diagram for multi-stage experiments (p. 719)
- **Example 13** — Using a tree diagram for experiments without replacement (p. 720)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 718)
- **Fluency:** Q1–4 (p. 721–722)
- **Problem-solving:** Q5–8 (p. 723)
- **Reasoning:** Q9–12 (p. 724)
- **Enrichment: Fermat and Pascal:** Q13 (p. 725)

Worked examples (study the method, then try the hurdles)

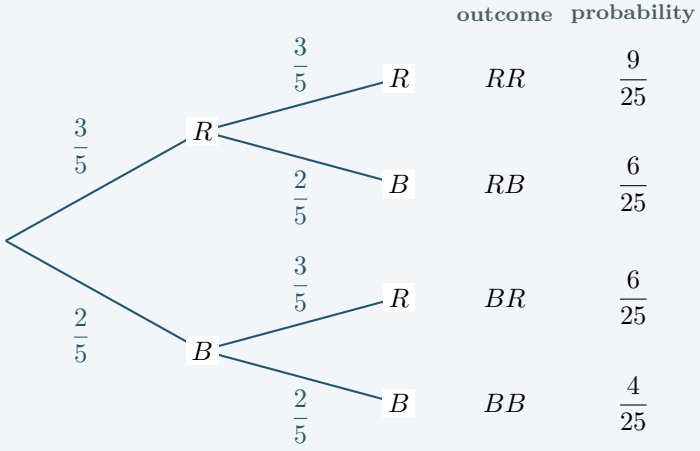
A tree diagram with replacement

A bag holds 3 red (R) and 2 blue (B) counters. A counter is drawn, its colour noted, and it is **replaced**; then a second counter is drawn. Find:

(a) $\Pr(RR)$

(b) $\Pr(\text{one of each colour})$

(c) $\Pr(\text{at least one blue})$.



Solution. Because the counter is replaced, the second-stage probabilities are the same as the first. **Multiply along** a path, then **add** the paths you need.

(a) $\Pr(RR) = \dfrac{3}{5} \times \dfrac{3}{5} = \dfrac{9}{25}$.

(b) One of each: $\Pr(RB) + \Pr(BR) = \dfrac{6}{25} + \dfrac{6}{25} = \dfrac{12}{25}$.

(c) At least one blue is everything except RR: $1 - \Pr(RR) = 1 - \dfrac{9}{25} = \dfrac{16}{25}$.

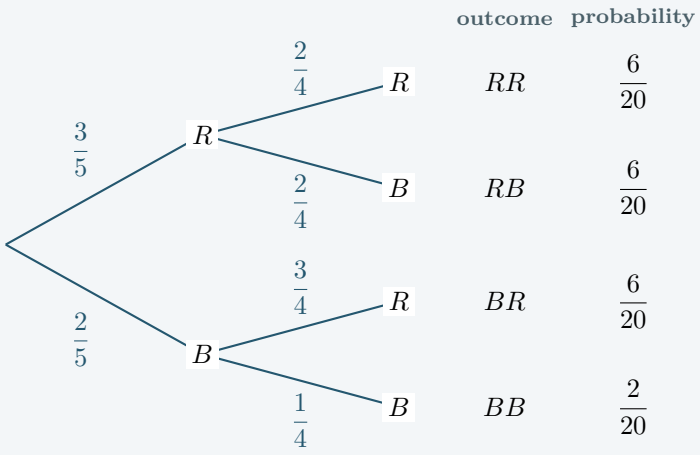
A tree diagram without replacement

The same bag (3 red, 2 blue) is used, but this time the first counter is **not** replaced. Find:

(a) $\Pr(RR)$

(b) $\Pr(\text{one of each colour})$

(c) $\Pr(BB)$.



Solution. After one counter is taken, only 4 remain, so every second-stage denominator is 4 — and the numerators depend on what was taken first.

(a) $\Pr(RR) = \dfrac{3}{5} \times \dfrac{2}{4} = \dfrac{6}{20} = \dfrac{3}{10}$.

(b) One of each: $\Pr(RB) + \Pr(BR) = \dfrac{6}{20} + \dfrac{6}{20} = \dfrac{12}{20} = \dfrac{3}{5}$.

(c) $\Pr(BB) = \dfrac{2}{5} \times \dfrac{1}{4} = \dfrac{2}{20} = \dfrac{1}{10}$.

Check: the outcomes cover everything, so $\dfrac{3}{10} + \dfrac{3}{5} + \dfrac{1}{10} = 1$.

Notes

Hurdle questions — Section 8F (show all working)

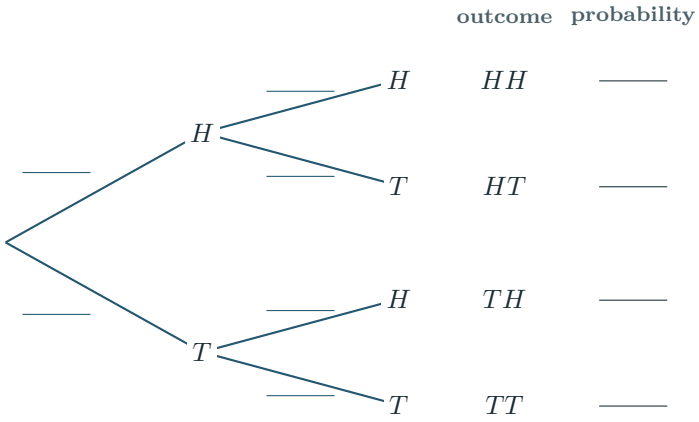
H1. A biased coin lands heads (H) with probability 0.6. It is tossed twice.

(a) Complete the tree diagram, including the probability of each outcome.

(b) Find Pr(HH).

(c) Find Pr(exactly one head).

(d) Find Pr(at least one tail).



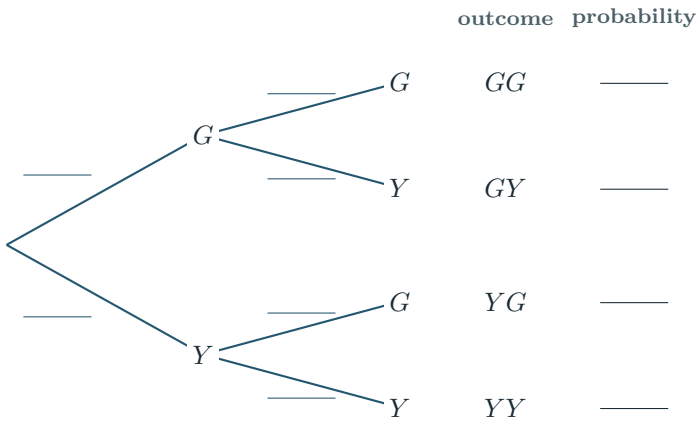
H2. A box holds 4 green (G) and 6 yellow (Y) tennis balls. Two balls are taken out **without replacement**.

(a) Complete the tree diagram.

(b) Find Pr(GG).

(c) Find Pr(both balls are the same colour).

(d) Find Pr(at least one green).



H3. The probability of rain on Monday is 0.3. If it rains on Monday, the probability of rain on Tuesday is 0.6. If it does not rain on Monday, the probability of rain on Tuesday is 0.2.

(a) Draw a tree diagram for the two days.

(b) Find the probability that it rains on both days.

(c) Find the probability that it rains on exactly one of the two days.

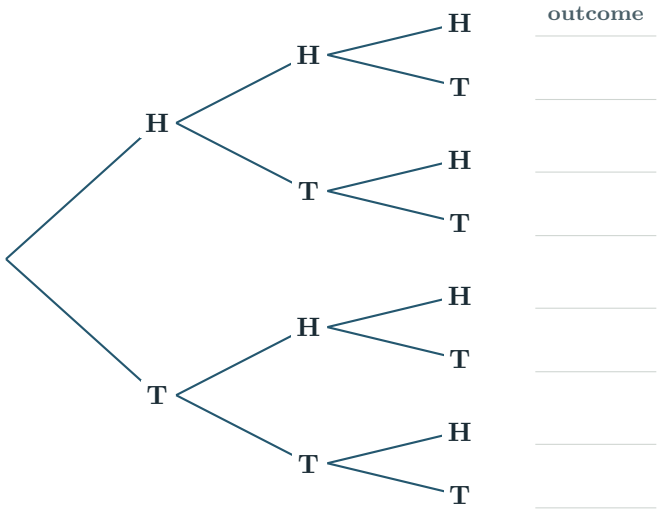
(d) Find the probability that it rains on Tuesday.

H4. A fair coin is tossed three times.

(a) Complete the tree diagram and list the eight outcomes.

(b) Find Pr(exactly two heads).

(c) Find Pr(at least one head).



H5. A bag holds 5 red and 3 white marbles. Two are drawn **without replacement**. A student writes $\Pr(RR) = \frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$.

(a) Explain the mistake.

(b) Find the correct value of $\Pr(RR)$.

8G Independent events CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 14** — Checking for independent events (p. 730)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 729)
- **Fluency:** Q1–4 (p. 731)
- **Problem-solving:** Q5–7 (p. 732)
- **Reasoning:** Q8–9 (p. 732)
- **Enrichment: Independence and the addition rule:** Q10–11 (p. 732)

Worked examples (study the method, then try the hurdles)

Testing for independence by multiplying

A number is chosen at random from 1 to 12. Let A be "an even number" and B be "a multiple of 3". Are A and B independent?

A and B are independent if $\Pr(A \cap B) = \Pr(A) \times \Pr(B)$ (equivalently $\Pr(A \mid B) = \Pr(A)$).

Solution. $\Pr(A) = \dfrac{6}{12} = \dfrac{1}{2}$ and $\Pr(B) = \dfrac{4}{12} = \dfrac{1}{3}$, so $\Pr(A) \times \Pr(B) = \dfrac{1}{6}$.
Both even *and* a multiple of 3: $\{6, 12\}$, so $\Pr(A \cap B) = \dfrac{2}{12} = \dfrac{1}{6}$. The two are equal, so A and B **are independent**.
Check with the other test: given $B = \{3, 6, 9, 12\}$, two of the four are even: $\Pr(A \mid B) = \dfrac{2}{4} = \dfrac{1}{2} = \Pr(A)$. Knowing B happened does not change the chance of A.

Testing for independence from a two-way table

Fifty students were surveyed about their year level and whether they wear glasses (G).

	G	G'	**Total**
Year 9	6	14	20
Year 10	9	21	30
Total	15	35	50

Is wearing glasses independent of being in Year 9?

Solution. $\Pr(G) = \dfrac{15}{50} = 0.3$ and $\Pr(G \mid \text{Year 9}) = \dfrac{6}{20} = 0.3$ (use the Year 9 row only).
These are equal, so wearing glasses **is independent** of being in Year 9. (The Year 10 row agrees: $\dfrac{9}{30} = 0.3$.)

Hurdle questions — Section 8G (show all working)

H1. A number is chosen at random from 1 to 20. Let A be "an even number" and B be "a multiple of 5".

(a) Find $\Pr(A)$, $\Pr(B)$ and $\Pr(A \cap B)$.

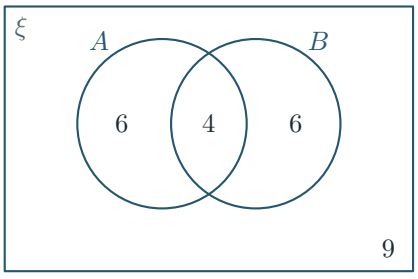
(b) Are A and B independent? Justify your answer.

H2. A number is chosen at random from 1 to 20. Let A be "an even number" and C be "a prime number".

(a) Find $\Pr(C)$ and $\Pr(A \cap C)$.

(b) Are A and C independent? Justify your answer.

H3. The Venn diagram shows the number of elements in events A and B. Decide whether A and B are independent.



H4. A and B are independent events with $\Pr(A) = 0.4$ and $\Pr(B) = 0.5$.

(a) Find $\Pr(A \cap B)$.

(b) Find $\Pr(A \cup B)$.

(c) Find $\Pr(A' \cap B')$.

H5. Eighty students were asked whether they walk to school (W).

	W	W'	**Total**
Year 7	12	28	40
Year 8	18	22	40
Total	30	50	80

Is walking to school independent of being in Year 7? Justify your answer.

8H Counting principles and factorial notation 10A

Textbook worked examples (cross-reference in your textbook)

- **Example 15** — Using the addition and multiplication principles (p. 734)
- **Example 16** — Evaluating factorials (p. 735)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 734)
- **Fluency:** Q1–4 (p. 735)
- **Problem-solving:** Q5–8 (p. 736)
- **Reasoning:** Q9–11 (p. 736)
- **Enrichment: Proofs with factorials:** Q12–14 (p. 736)

Worked examples (study the method, then try the hurdles)

"Or" adds, "and" multiplies

A café sells 4 hot drinks, 5 cold drinks and 3 cakes. How many ways can you choose:

(a) one drink

(b) one drink and one cake

(c) one hot drink and one cold drink?

Solution.

(a) One drink is a hot drink *or* a cold drink: **addition principle**, $4 + 5 = 9$ ways.

(b) A drink *and* a cake: **multiplication principle**, $9 \times 3 = 27$ ways (each of the 9 drinks can go with each of the 3 cakes).

$$\underset{\text{drink}}{\boxed{9}} \times \underset{\text{cake}}{\boxed{3}}$$

(c) A hot drink *and* a cold drink: $4 \times 5 = 20$ ways.

Notes

Working with factorials

Evaluate without a calculator:

(a) $6!$

(b) $\dfrac{8!}{6!}$

(c) $\dfrac{10!}{7!\,3!}$

(d) Simplify $\dfrac{(n+1)!}{n!}$.

Solution. $n! = n \times (n-1) \times \cdots \times 2 \times 1$, and $0! = 1$.

(a) $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

(b) $\dfrac{8!}{6!} = \dfrac{8 \times 7 \times \cancel{6!}}{\cancel{6!}} = 8 \times 7 = 56$

(c) $\dfrac{10!}{7!\,3!} = \dfrac{10 \times 9 \times 8 \times \cancel{7!}}{\cancel{7!} \times 3 \times 2 \times 1} = \dfrac{720}{6} = 120$

(d) $\dfrac{(n+1)!}{n!} = \dfrac{(n+1) \times \cancel{n!}}{\cancel{n!}} = n+1$

Tip: write the larger factorial down to the smaller one, then cancel.

Notes

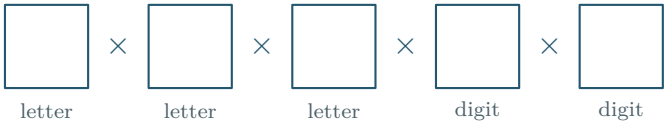
Hurdle questions — Section 8H (show all working)

H1. A school offers 5 language electives and 7 arts electives. How many ways can a student choose:

(a) one elective from either list

(b) one language elective and one arts elective?

H2. A number plate has 3 letters (A to Z) followed by 2 digits (0 to 9). Letters and digits may repeat. How many different number plates are possible?



H3. Evaluate without a calculator:

(a) $5!$

(b) $\dfrac{9!}{7!}$

(c) $\dfrac{7!}{4!\,3!}$

(d) $0!$

H4. Simplify:

(a) $\dfrac{n!}{(n-1)!}$

(b) $\dfrac{(n+2)!}{n!}$

H5. A code has 4 characters. Each character is a digit (0 to 9) or one of the letters A to F. How many codes are possible if:

(a) characters may repeat

(b) no character may be used more than once?

8I Arrangements 10A

Textbook worked examples (cross-reference in your textbook)

- **Example 17** — Finding the total number of arrangements (p. 739)
- **Example 18** — Finding arrangements with restrictions (p. 739)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–4 (p. 738)
- **Fluency:** Q1–5 (p. 740)
- **Problem-solving:** Q6–9 (p. 740)
- **Reasoning:** Q10–12 (p. 741)
- **Enrichment: How many numbers?:** Q13–15 (p. 741)

Worked examples (study the method, then try the hurdles)

Counting arrangements with boxes

(a) In how many ways can 6 different books be arranged on a shelf?

(b) Eight runners race. In how many ways can first, second and third place be filled?

Solution. Draw one box per position and fill it with the number of choices left.

(a) $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 6! = 720$.

(b) 8 runners can come first, then 7 are left for second, then 6 for third: $8 \times 7 \times 6 = 336$.

8	$\times$	7	$\times$	6
1st		2nd		3rd

This is ${}^{8}P_{3} = \dfrac{8!}{(8-3)!} = \dfrac{8!}{5!} = 336$.

Notes

Arrangements with restrictions

The letters of the word **PRIME** are arranged in a row. How many arrangements:

(a) are there altogether

(b) begin with P

(c) begin and end with a vowel

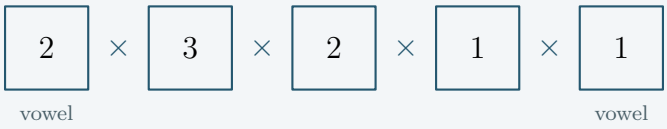
(d) have the two vowels next to each other?

Solution. Fill the restricted positions first.

(a) No restrictions: $5! = 120$.

(b) The first box must be P (1 way); the other 4 letters fill the rest: $1 \times 4! = 24$.

(c) First box: 2 vowels (I or E); last box: the 1 vowel left; the 3 consonants fill the middle: $2 \times 3 \times 2 \times 1 \times 1 = 12$.



(d) Glue I and E into one block: arrange 4 items (4!), then arrange I and E inside the block (2!): $4! \times 2! = 48$.

Notes

Hurdle questions — Section 8I (show all working)

H1. In how many ways can 7 people stand in a queue?

H2. A club has 10 members. In how many ways can a president, a vice-president and a secretary be chosen? Write your answer using ${}^{n}P_{r}$ notation as well.



H3. The letters of the word **FRACTION** are arranged in a row. How many arrangements:

(a) are there altogether

(b) begin with F

(c) begin with a vowel?

H4. Five friends, including Ana and Ben, sit in a row of five seats. In how many ways can they sit if:

(a) Ana and Ben must sit together

(b) Ana and Ben must **not** sit together?

H5. Three-digit numbers are made from the digits $1, 2, 3, 4, 5, 6$, with no digit repeated. How many of these numbers:

(a) are there altogether

(b) are even

(c) are greater than 400?

8J Selections 10A

Textbook worked examples (cross-reference in your textbook)

- **Example 19** — Evaluating ${}^nC_r$ or $\binom{n}{r}$ (p. 743)
- **Example 20** — Finding the number of possible selections (p. 744)
- **Example 21** — Making selections from two groups (p. 744)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 742–743)
- **Fluency:** Q1–5 (p. 744–745)
- **Problem-solving:** Q6–8 (p. 745)
- **Reasoning:** Q9–11 (p. 745)
- **Enrichment: Choosing any number of options:** Q12–14 (p. 745)

Worked examples (study the method, then try the hurdles)

Evaluating ${}^nC_r$

(a) Evaluate ${}^7C_3$.

(b) Evaluate $\binom{10}{2}$.

(c) Explain why ${}^7C_3 = {}^7C_4$.

$$ {}^nC_r = \binom{n}{r} = \frac{n!}{r!\,(n-r)!} $$

Solution.

(a) ${}^7C_3 = \dfrac{7!}{3!\,4!} = \dfrac{7\times6\times5}{3\times2\times1} = \dfrac{210}{6} = 35$

(b) $\binom{10}{2} = \dfrac{10!}{2!\,8!} = \dfrac{10\times9}{2\times1} = 45$

(c) ${}^7C_4 = \dfrac{7!}{4!\,3!}$, which is the same expression as ${}^7C_3$. Choosing the 3 items to take is the same as choosing the 4 items to leave behind.

Notes

Selections from one group and from two groups

(a) A committee of 3 is chosen from 9 people. How many committees are possible?

(b) A team of 2 teachers and 3 students is chosen from 5 teachers and 6 students. How many teams are possible?

Solution. In a committee the order does not matter, so this is a **selection**.

(a) ${}^9C_3 = \dfrac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84.$

(b) Choose the teachers *and* the students, then multiply: ${}^5C_2 \times {}^6C_3 = 10 \times 20 = 200.$

Notes

Hurdle questions — Section 8J (show all working)

H1. Evaluate without a calculator:

(a) ${}^{6}C_{2}$

(b) $\binom{8}{3}$

(c) ${}^{12}C_{1}$

(d) ${}^{5}C_{5}$

H2. A pizza shop offers 10 toppings. In how many ways can 4 different toppings be chosen?

H3. A coach chooses a team of 5 from a squad of 12 players.

(a) How many teams are possible?

(b) How many of these teams include the captain?

H4. A group of 3 boys and 2 girls is chosen from 7 boys and 5 girls. How many groups are possible?

H5. Two letters are chosen from A, B, C and D.

(a) List every **arrangement** (order matters) of two letters. How many are there?

(b) List every **selection** (order does not matter) of two letters. How many are there?

(c) Explain how the two answers are connected, using 2!.

End of Chapter 8 hurdle booklet — check your working against the worked examples and ask in class if any hurdle question is unclear.